



Patient-reported treatment outcomes in ME/CFS and long COVID

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Contributed by Ronald Davis; received December 24, 2024; accepted April 7, 2025; reviewed by Lucinda Bateman and Lewis E. Kazis

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) and long COVID are persistent multisystem illnesses affecting many patients. With no known effective FDA-approved treatments for either condition, patient-reported outcomes of treatments may prove helpful in identifying management strategies that can improve patient care and generate new avenues for research. Here, we present the results of an ME/CFS and long COVID treatment survey with responses from 3,925 patients. We assess the experiences of these patients with more than 150 treatments in conjunction with their demographics, symptoms, and comorbidities. Treatments with the greatest perceived benefits are identified. Patients with each condition who participated in the study shared similar symptom profiles, including all the core symptoms of ME/CFS, e.g., 89.7% of ME/CFS and 79.4% of long COVID reported postexertional malaise (PEM). Furthermore, treatment responses between these two patient groups were significantly correlated ($R^2 = 0.68$). Patient subgroups, characterized by distinct symptom profiles and comorbidities, exhibited increased responses to specific treatments, e.g., a POTS-dominant cluster benefiting from autonomic modulators and a cognitive-dysfunction cluster from CNS stimulants. This study underscores the symptomatic and therapeutic similarities between ME/CFS and long COVID and highlights the commonalities and nuanced complexities of infection-associated chronic diseases and related conditions. While this study does not provide recommendations for specific therapies, in the absence of approved treatments, insights from patient-reported experiences provide urgently needed real-world evidence for developing targeted patient care therapies and future clinical trials.

ME/CFS | long COVID | treatments | patient-reported outcomes | real-world evidence

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) and long COVID are both debilitating, multisystem illnesses that profoundly impact millions of people worldwide (1–5). Long COVID alone currently affects 17.9% of adults in the United States (Sept 2024) (6), with many of these patients meeting the diagnostic criteria for ME/CFS (3, 7–9). Before the pandemic, ME/CFS was estimated to affect up to 3.3 million people in the United States (1, 10–12), but the surge in long COVID cases is increasing these numbers significantly. The large socioeconomic burden of these conditions will continue to escalate if effective treatments are not identified (12, 13).

Increasing evidence suggests that long COVID and the majority of ME/CFS cases are infection-associated chronic conditions in which initial infections trigger downstream pathological mechanisms, leading to chronic, multisystem dysfunction (2, 9, 14). Despite the diverse symptoms in individual patients, studies have shown that many long COVID and ME/CFS sufferers share hallmark symptoms such as fatigue, postexertional malaise (PEM), unrefreshing sleep, orthostatic intolerance (OI), and cognitive dysfunction. These symptoms align with the 2015 Institute of Medicine (IOM) diagnostic criteria for ME/CFS (2, 3, 8, 14–17). Additional symptoms and comorbidities, such as pain (e.g., migraines, neuropathy) and mast cell activation syndrome (MCAS), are also prevalent in both conditions. Furthermore, both conditions occur more frequently in females, with the average age of onset typically in the 30 s or 40 s (4, 10, 11, 18, 19). These similarities suggest potential links in the underlying pathophysiology of ME/CFS and long COVID.

Patients with ME/CFS and long COVID frequently struggle to access specialized medical care (20, 21). Primary care providers and other health professionals often lack the necessary expertise to diagnose and manage ME/CFS or long COVID effectively, leading to significant delays in appropriate treatments, which negatively impact patients' physical and emotional well-being. Even when patients do access appropriate medical care, treatment options are limited: With no known cure for long COVID or ME/CFS, patients must rely on palliative treatments and less accessible off-label medications (22–24). Due to the variety of symptoms and comorbidities found in these conditions, treatments must be uniquely tailored to individual needs, often on a trial-and-error basis. Several systematic reviews summarize the existing clinical evidence of therapies in ME/CFS (25–27). Early

Significance

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) and long COVID are debilitating conditions that currently lack FDA-approved treatments. This study analyzes patient-reported outcomes from over 3,900 individuals, identifying treatments perceived as beneficial and uncovering symptom-based patient subgroups with distinct responses to therapies. Notably, there is significant overlap in the symptom profiles and treatment responses between ME/CFS and long COVID, suggesting that they may share underlying mechanisms. These findings offer valuable real-world insights for patients and their healthcare providers and help identify promising candidates for clinical trials, addressing an urgent need for effective therapies in these chronic illnesses.

Author contributions: M.E., R.W.D., and W.X. designed research; M.E., P.L., B.M., and W.X. performed research; M.E., P.L., B.M., J.B., R.W.D., and W.X. analyzed data; R.W.D. and W.X. supervised the project; and M.E., P.L., B.M., J.B., R.W.D., and W.X. wrote the paper.

Reviewers: L.B., Bateman Horne Center (BHC); and L.E.K., Boston University School of Public Health.

Competing interest statement: Dr. Davis and Dr. Xiao serve on the Scientific Advisory Board of the Open Medicine Foundation, a non-profit charity that supports collaborative medical research for ME/CFS and other complex chronic diseases.

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This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2426874122/-DCSupplemental>.

Published July 8, 2025.

reports suggest that many of the treatments—such as low-dose naltrexone (LDN), pyridostigmine (Mestinon), beta-blockers, immunoglobulin therapy (IVIG), and CNS stimulants—may also help improve corresponding symptoms in long COVID (3, 24, 28, 29). Additionally, long COVID studies investigating platelet hyperactivation and other prothrombotic changes (30–32) have renewed interest in the use of antiplatelets, anticoagulants, and fibrinolytic supplements in ME/CFS (33–35). As we await further clinical trials, real-world patient reports via surveys can provide quick and direct insights into which therapeutic interventions may or may not be beneficial. Such information has long impacted drug development (36–38) and is particularly important in the management of chronic diseases (39–41).

Here, we present a comprehensive study of treatment outcomes in 3,925 patients with ME/CFS and long COVID (*Materials and methods*). The TREATME survey examined over 150 supplements, over-the-counter medications, prescription medications, and non-pharmacological interventions and their perceived effectiveness in managing these debilitating conditions. These treatments were selected based on pharmacological and nonpharmacological therapies currently used in ME/CFS (25–27), various pertinent studies, and conversations with members of the patient community. While this work does not make specific treatment recommendations, it offers important insights into the huge variability in treatments tested and their general outcomes. As the data rely on self-reported inputs, there are inherent uncertainties in the individual response, but at a group level, it should provide a solid base for interpretations. By assessing the symptoms and comorbidities of patients along with their treatment experiences, we identified therapies with the greatest perceived benefits and the core symptoms they improved. Patient subgroups with distinct symptom profiles also were identified, showing increased responses to specific treatments. This research establishes a basis for utilizing real-world patient data to better identify and understand potential therapies for ME/CFS and long COVID.

Results

Demographics and Disease Duration of Patients in the Study.

3,925 patients responded to the survey and answered questions on demographics, symptom profiles, and comorbidities, including 2,125 patients with ME/CFS and 1,800 patients with long COVID. As shown in Table 1, 80.0% of the long COVID patients (1,440/1,800) and 87.5% of the ME/CFS patients (1,859/2,125) reported that their diagnoses were made formally, while the rest reported that their diagnoses were strongly suspected either by the patients themselves or their doctors. Altogether, 76.4% of long COVID patients and 82.8% of ME/CFS patients were female, and 83.7% of long COVID patients and 77.2% of ME/CFS patients were between 30 and 65 y old. The average ages of the patients were 47.5 ± 14.9 and 44.1 ± 12.9 for ME/CFS and long COVID, respectively. These similarities in age and sex breakdown between patient groups roughly align with those reported elsewhere, suggesting that survey respondents are broadly representative of these patient populations.

Given the long history of ME/CFS and the recent emergence of long COVID, the duration of illness for patients differed widely between the groups. The average disease durations (years) were 12.7 ± 8.5 for ME/CFS and 1.6 ± 0.9 for long COVID, respectively. While some ME/CFS patients also reported illness durations exceeding 20 y, the length reported by long COVID patients was naturally capped at 3 y (2020–2023). Both groups reported significantly reduced patient capacity levels (Table 1), and ME/CFS patients had, on average, a lower percentage of their preillness capacity compared

Table 1. Characteristics of ME/CFS and long COVID patients in the study

Characteristic	No. (%)	
	ME/CFS (n = 2,125)	Long COVID (n = 1,800)
Sex at birth		
Female	1,759 (82.8)	1,376 (76.4)
Male	353 (16.6)	419 (23.3)
Prefer not to say	13 (0.6)	5 (0.3)
Age, year		
5 to 11	1 (0.0)	5 (0.3)
12 to 17	15 (0.7)	18 (1.0)
18 to 29	231 (10.9)	197 (10.9)
30 to 39	430 (20.2)	437 (24.3)
40 to 49	533 (25.1)	589 (32.7)
50 to 65	677 (31.9)	481 (26.7)
66+	235 (11.1)	72 (4.0)
Not Available	3 (0.1)	1 (0.1)
Mean $\pm$ SD	47.5 ± 14.9	44.1 ± 12.9
Diagnosis Status		
Formally diagnosed	1,859 (87.5)	1,440 (80.0)
Self-diagnosed	266 (12.5)	360 (20.0)
Disease Duration, year		
<1	37 (1.7)	568 (31.6)
1 ~ 3	283 (13.3)	1,165 (64.7)
4 ~ 20	1,259 (59.2)	0 (0.0)
20+	546 (25.7)	0 (0.0)
Not Available	0 (0.0)	67 (3.7)
Mean $\pm$ SD	12.7 ± 8.5	1.6 ± 0.9
Patient Capacity Level*		
Fit and well for ≥ 1 mo (100%)	8 (0.4)	20 (1.1)
Generally well (~90%)	30 (1.4)	78 (4.3)
Mild (70 to 90%)	144 (6.8)	247 (13.7)
Mild to moderate (50 to 70%)	294 (13.8)	390 (21.7)
Moderate (40 to 50%)	530 (24.9)	498 (27.7)
Moderate to severe (25 to 40%)	639 (30.1)	358 (19.9)
Severe (15 to 25%)	356 (16.8)	159 (8.8)
Very severe (<15%)	78 (3.7)	23 (1.3)
Not Available	46 (2.2)	27 (1.5)
Mean $\pm$ SD	41.0 ± 18.5	50.6 ± 20.4

*SI Appendix, Table S1.

to long COVID patients ($41.0 \pm 18.5\%$ vs $50.6 \pm 20.4\%$). More ME/CFS patients (20.4%) also reported severe (15–25% capacity) to very severe illness (<15% capacity), compared to those with long COVID (10.1%). Additionally, only 8.6% of ME/CFS patients reported their capacity as “mild” (70 to 90% capacity) or higher, compared to 19.1% of long COVID patients.

ME/CFS and Long COVID Patients Share Similar Symptom Profiles.

Patients reported the most troubling symptoms experienced over the course of their illness (SI Appendix, Table S2). Fig. 1 shows the top symptoms ranked by their prevalence among respondents with long COVID. The top five symptoms are fatigue (95.6% of ME/CFS patients, 88.3% of long COVID patients), PEM (89.7%, 79.4%), brain fog (80.1%, 72.3%), unrefreshing sleep (74.5%,

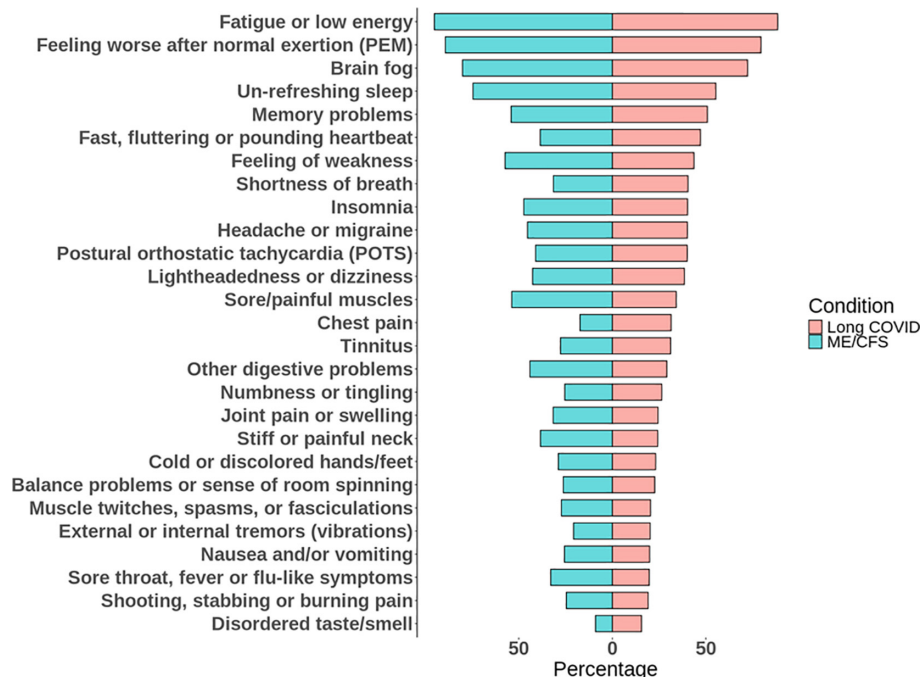


Fig. 1. Most troubling symptoms in ME/CFS and long COVID patients. The bar graph shows the frequency of symptoms reported by patients with ME/CFS (blue) and long COVID (red), ordered based on the frequency in long COVID patients.

55.3%), and memory problems (54.1%, 50.8%). Both groups also reported high rates of postural orthostatic tachycardia syndrome (POTS; 41.0%, 40.0%), along with other OI-related symptoms like fast, fluttering or pounding heartbeat (38.5%, 47.1%) and lightheadedness and dizziness (42.6%, 38.5%). Notably, this symptom profile aligns with the IOM ME/CFS criteria. In addition, both ME/CFS and long COVID patients had similar rates of insomnia (47.3%, 40.2%), headache or migraine (45.3%, 40.1%), and numbness and tingling (25.4%, 26.4%). In contrast, a lower percentage of ME/CFS patients experienced frequent shortness of breath (dyspnea; 31.5%, 40.4%), chest pain (17.3%, 31.4%), and disordered taste/smell (9.0%, 15.6%), possibly due to the unique respiratory, cardiovascular, and sensory impacts of acute SARS-CoV-2 infection. On the other hand, ME/CFS patients had a higher rate of sore/painful muscles (53.7%, 34.1%).

ME/CFS and long COVID patients also had similar comorbidities, including anxiety and depression (56.6% in ME/CFS, 51.1% in long COVID), POTS (51.9%, 48.7%), migraine (49.1%, 41.1%), other dysautonomia (37.5%, 38.9%), mast cell activation syndrome (MCAS; 31.8%, 29.1%), Ehlers–Danlos syndrome (EDS)/joint hypermobility (40.2%, 27.2%), and ADD/ADHD (23.8%, 23.7%) (*SI Appendix, Fig. S1*). To evaluate how comorbidities might affect symptoms that patients experienced, we further evaluated their correlations (*SI Appendix, Fig. S2*). Most of the significant correlations are expected, such as POTS/OI as comorbidities and POTS as a symptom, and Fibromyalgia as a comorbidity and pain symptoms as expected correlations. These findings are consistent with other reports suggesting overlapping pathologies of ME/CFS and long COVID (3, 7, 8, 15, 16, 28, 42, 43), which led us to hypothesize that ME/CFS and long COVID patients may show similar responses to certain treatments.

Treatments Identified with the Greatest Perceived Benefit Reported by Patients. The survey included over 440 questions on the use and perceived efficacy of over 150 different interventions. Patients reported how they felt each treatment influenced their overall condition, choosing from seven options: much better,

moderately better, slightly better, about the same/unchanged, slightly worse, moderately worse, or much worse. Additionally, patients were asked to choose from those same seven options to evaluate a select treatment's effects on their most troubling symptoms. The detailed results are shown in Supplementary Data.

Reported effect on the overall condition was summarized using a Net Assessment Score (NAS) (*Materials and Methods*). Each treatment evaluated by 20 or more patients was assigned a NAS. The NAS of each treatment was compared to the NAS provided by patients for an oral, nonliposomal Vitamin C supplement—which, in the absence of a placebo, served as a reference—using the false discovery rate adjusted *P*-value (adj.*P*) (Fig. 2). The results of this analysis for each treatment are in *SI Appendix, Table S3*.

Fig. 2*A* shows the top twenty treatment groups with the greatest perceived benefit reported by patients. For example, fluids/electrolytes demonstrated significantly higher perceived benefit over the reference (adj.*P* = 1.0e–171) and a NAS of 68.6% based on 3,053 responses. Other treatment groups with the most positive NAS include pacing (75.2%), compression stockings (62%), intravenous/subcutaneous immunoglobulin (IgG; 58.2%), maraviroc (56.9%), manual lymphatic drainage (55.7%), antihistamines (51.6%), nattokinase/lumbrokinase (NK/LK; 49.9%), low-dose naltrexone (LDN; 49.4%), beta blocker or ivabradine (47.1%), vagal nerve stimulation (45.5%), Rx anticoagulants/Rx antiplatelets (43.7%), melatonin (43.3%), palmitoylethanolamide (PEA; 41.5%), ADHD stimulants (41.6%), and Mestinon (41%).

Several individual treatments in these treatment groups received high percentages (>40%) of responses indicating “much better” and “moderately better,” including IV saline, ivabradine, IgG, heparin (UFH and LMWH), and maraviroc (*SI Appendix, Table S3*). However, a relatively small sample size (<100 patients) evaluated IgG, maraviroc, or heparin. Also notably, the combination of H1 and H2 histamine receptor antagonists (H1RA + H2RA) had the greatest positive impact (NAS of 63.4%) among treatment regimens for MCAS symptoms, and Ivabradine (66.8%) had a significantly higher positive impact than beta blockers. In addition, PEA formulations with enhanced bioavailability had a

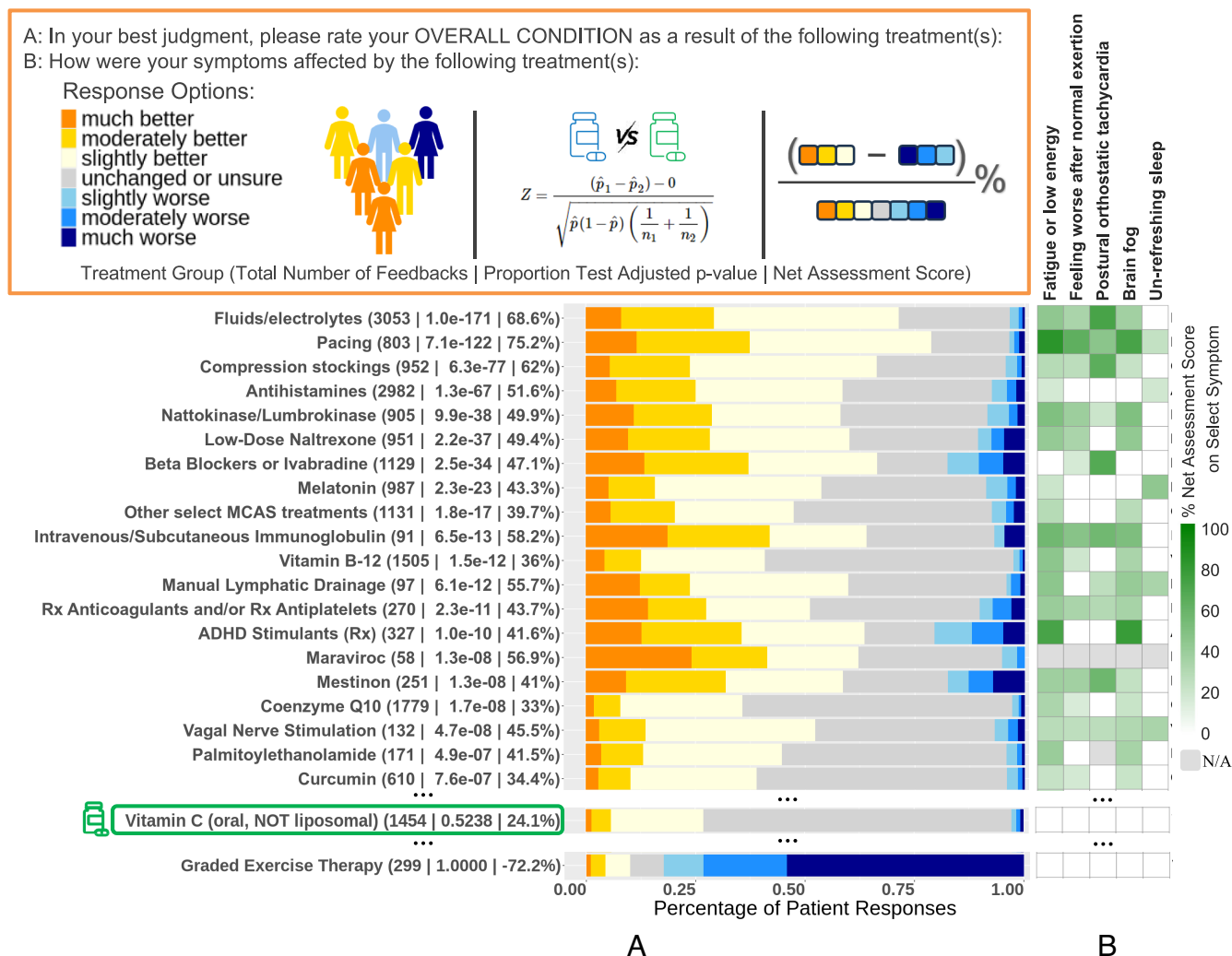


Fig. 2. Patient-reported outcomes of treatments in ME/CFS and long COVID. (A) The leading treatments and categories that contributed to improving patients' overall conditions as reported in self-assessments. The data within parentheses next to each treatment indicate the number of responses provided on the treatment, the false discovery rate adjusted *P*-value compared to the reference treatment, and the Net Assessment Score (NAS). The stacked bar plots show the percentages of patient responses related to the perceived treatment effect of each treatment, spanning from "much worse" to "much better". Orange shading denotes the percentage of patients who reported a positive treatment effect—with darker orange indicating a stronger positive effect—whereas blue shading indicates adverse side effects, with darker blue representing more significant negative effects. Vitamin C (oral, nonliposomal) was used as the reference treatment. Of all the treatments investigated, Graded exercise therapy (GET) received the lowest NAS. (B) Impact of treatments on patient symptoms as reported in self-assessments. For each of the core symptoms, shown in a gradient scale is the proportion of respondents of a treatment who also reported the impact of the treatment improving the specific symptom.

higher positive response (56.8%), B12 injections (47.1%) significantly outperformed oral B12 (30.5%), and higher doses of coenzyme significantly outperformed lower dosages (e.g., 50.7% for >200 mg/d versus 26.7% for 50 to 100 mg/d).

It is also important to note that while the majority of patients reported benefits from medications such as beta-blockers, ADHD stimulants, and Mestinon, a smaller group (17 to 21%) indicated that these medications worsened their condition. This highlights patients' diverse experiences with treatments.

Examining other treatments with responses from at least 100 patients (*SI Appendix, Table S3*), N-acetyl cysteine (NAC >600 mg/d, 45.8%), (Acetyl)-L-carnitine (≥500 mg/d, 41.7%), glutathione injection (42.4%), EPA or icosapent ethyl (44.1%), and corticosteroids (38.4%) also exhibited significant positive effects. Conversely, a considerable number of other treatments did not show overall significance compared to the reference. Some of these, including many neuropsychiatric drugs (e.g., SSRIs, SNRIs, amitriptyline & other TCAs, trazodone or nefazodone,

gabapentinoids, (ar)modafinil, and Abilify), along with dihydropyridines (a subset of calcium channel blockers), fludrocortisone, midodrine, and cognitive behavioral therapy (CBT), all had over 20% of patients responding negatively. However, notably, trazodone/nefazodone, pregabalin, gabapentin, Abilify ≤2 mg (low-dose), and midodrine also had more than 50% of positive responses. This again underscores the diverse responses of patients to treatments. Of all treatments investigated, graded exercise therapy (GET) received the lowest NAS (−72.2%) by far, with the vast majority of polled patients reporting harms and almost none reporting benefit (Fig. 2A).

Treatments Impact Core Disease Symptoms Differently. As stated earlier, our survey asked patients to evaluate a treatment's effects not only on their overall condition but also on their most troubling symptoms. The five symptoms we included in our primary analysis were selected to align with IOM's core diagnostic criteria for ME/CFS: 1) fatigue or low energy, 2) feeling worse

after normal exertion (PEM), 3) POTS, 4) brain fog, and 5) unrefreshing sleep. These symptoms were also among the topmost troubling symptoms selected by survey participants in both ME/CFS and long COVID. Patients were asked to report the extent to which each of their selected symptoms was impacted by each treatment, again selecting from seven options (e.g., “much better,” “slightly worse,” et al.) To investigate the effects of the top 20 treatment groups on these five core symptoms, we calculated symptom-specific NASs, which then were compared to the NAS for the reference, vitamin C, to determine which were significantly different (*SI Appendix, Table S4*). Fig. 2*B* shows the frequencies of a treatment that significantly improved each symptom compared to vitamin C (adj. *P*-value < 0.05).

Each treatment exerted varying effects on symptoms, underscoring the importance of managing these conditions based on patients’ individual needs. For example, patients reported that pacing helped frequently with fatigue (82.7%), PEM (62.6%), brain fog (71.2%), and POTS (45.5%). Similarly, IgG improved these symptoms frequently: fatigue (55.4%), PEM (46.7%), POTS (55.8%), and brain fog (50.8%). Fluids/electrolytes most frequently helped POTS (71.7%), and to a lesser extent, fatigue or low energy (43.2%), PEM (33.2%), and brain fog (36.5%). As expected, compression stockings and Mestinon most frequently improved symptoms of POTS (63.7% and 56.9%, respectively), while beta blockers and ivabradine primarily alleviated POTS symptoms (66.0%). LDN, on the other hand, did not improve POTS but significantly improved three other core symptoms: fatigue or low energy (41.5%), PEM (33.2%), and brain fog (42.3%). Interestingly, NK/LK similarly improved these three symptoms (49.7%, 38.2%, and 49.5%, respectively), and also showed some benefits in POTS (20.3%). While Rx anticoagulants/antiplatelets also improved these four core symptoms (30 to 40%), an additional symptom improved more significantly: shortness of breath (52.6%). As expected, ADHD stimulants significantly alleviated brain fog (77.1%) and general fatigue (71.7%) but did not improve PEM or POTS (−1.5% and −3.0%, respectively). Among the treatments surveyed, only melatonin significantly improved unrefreshing sleep (43.0%). In addition, patients with specific comorbidities (e.g., OI) were often correlated with using treatments that contributed to improving the symptoms (e.g., Mestinon) (*SI Appendix, Fig. S3*).

Responses to Treatments are Similar Between ME/CFS and long COVID. To compare the treatment response between the two conditions, we performed analyses of ME/CFS and long COVID patients separately (*Materials and Methods*). As shown in Fig. 3, the NASs of treatments were highly correlated between ME/CFS and long COVID patients ($R^2 = 0.68$), indicating similarities in treatment responses. This correlation was particularly pronounced for treatments reported by larger numbers of ME/CFS and long COVID patients. For complete results, see *SI Appendix, Table S5*. However, overall, long COVID patients often reported more positive response, i.e., higher NAS, to many treatments. Still, only two treatment groups—midodrine, benfotiamine, or thiamine tetrahydrofurfuryl disulfide (TTFD)—showed significantly different responses between the two conditions (adj. *P* < 0.05 and fold change > 1.25).

To further explore the differences seen between ME/CFS and long COVID patients, we identified significant predictors of treatment effectiveness, i.e., the NAS, from disease diagnosis (ME/CFS vs. long COVID), severity (patient capacity level), and demographics of patients (*SI Appendix, Fig. S4*). Disease severity, i.e., patient capacity level, was by far the most significant predictor of treatment effectiveness. Of the remaining variables, sex, disease

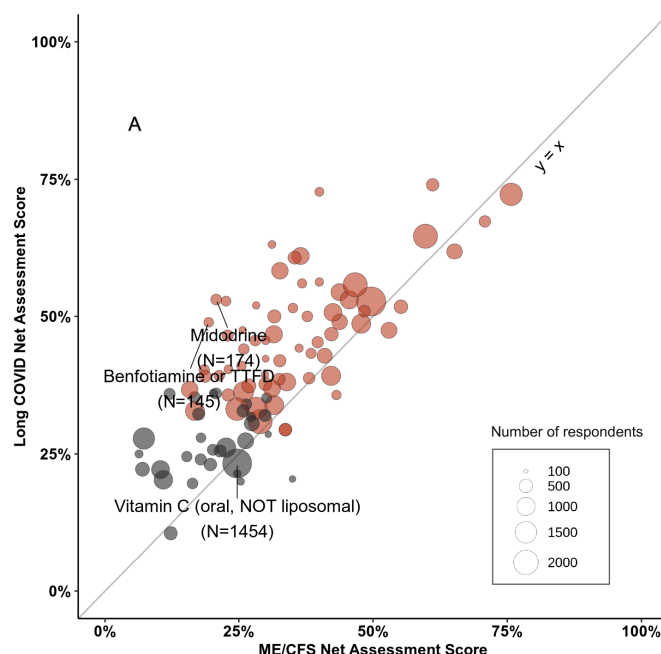


Fig. 3. Comparison of treatment effectiveness reported in ME/CFS and long COVID. Shown is the scatter plot of the NAS of treatments in ME/CFS (x-axis) compared to long COVID (y-axis), with each circle symbolizing a treatment. Red circles represent treatments that showed significance when compared to the reference (Vitamin C; oral, NOT liposomal), and gray circles indicate otherwise. The size of a circle represents the total number of respondents to the treatment.

duration, diagnosis status, and age all show a greater influence on the treatment outcome than did the patient’s diagnosis of ME/CFS or long COVID. Therefore, the diagnosis of long COVID and ME/CFS alone does not impact responses to treatments significantly.

Subgroups of Patients Had Distinct Profiles of Symptoms and Comorbidities. Since the survey results show that treatments impact disease symptoms differently, we further clustered the symptoms and comorbidities of the survey respondents (*Materials and Methods*).

Patients formed four clusters with distinct profiles (Fig. 4*A*): Cluster 1: Multisystemic Symptomatology, which was characterized by the highest rates of nearly all symptoms and comorbidities than those reported for other clusters. Cluster 2: POTS-Dominant Presentation, which was primarily marked by the highest rates of POTS among all clusters. Cluster 3: Cognitive and Sleep Dysfunction with Increased Pain, but markedly low rates of POTS symptoms. Cluster 4: Milder Symptomatology, where patients reported the lowest rates of nearly all symptoms and comorbidities. The characteristics of these clusters are further analyzed below.

Cluster 1: Multisystemic Symptomatology. Patients in Cluster 1 reported the lowest functional capacity level ($39.1 \pm 17.6\%$, *SI Appendix, Fig. S5*). Patients in this cluster showed higher rates of nearly every queried symptom and comorbidity than other clusters, and as expected, the most prevalent symptoms are the core symptoms of the disorder: fatigue (98.9%), PEM (97.4%), brain fog (93.6%), unrefreshing sleep (95.2%), memory problems (82.4%), feeling of weakness (87.3%), sore/painful muscles (87.0%), insomnia (76.1%), and headache/migraine (79.8%). The exceptions to this trend are POTS symptoms (69.0%) and POTS diagnosis (71.0%).

Cluster 2: POTS-Dominant Presentation. The functional capacity level of patients in Cluster 2 is $40.5\% \pm 18.1\%$. Patients in this cluster exhibited the highest rates of POTS as a primary

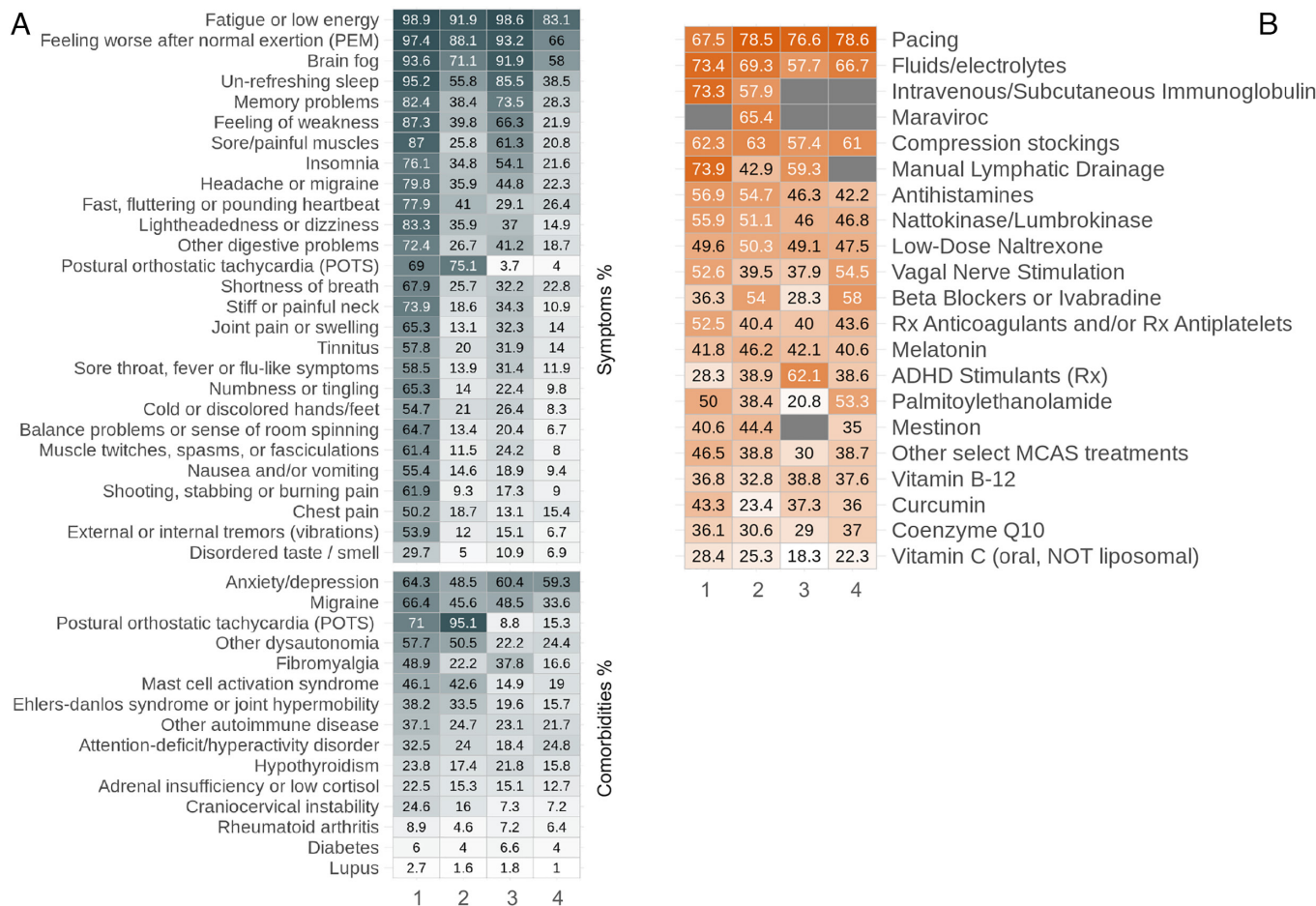


Fig. 4. Patient subgroups with distinct symptom profiles and treatment efficacies. (A) The heatmap displays the percentages of reported symptoms and comorbidities within each patient cluster. Cluster 1 had the highest prevalence of symptoms and comorbidities, Cluster 2 was predominated by postural orthostatic tachycardia syndrome (POTS) issues, Cluster 3 was characterized by cognitive symptoms like brain fog and sleep disturbances, and an increased reporting of pain, and Cluster 4 had the mildest symptoms. (B) The heatmap shows the comparative effectiveness of treatments across the four distinct patient clusters. The treatments listed are consistent with those reported in Fig. 2, and the NAS for each treatment in patients of each cluster is shown. The colors indicate treatment efficacies within each cluster, with warmer colors (reds and oranges) representing higher positive impact, and cooler colors (blues) representing lower positive impact. Gray denotes insufficient feedback (<20) for reliable NAS calculation.

symptom (75.1%) and comorbidity (95.1%). In contrast, compared to Cluster 1 and Cluster 3, these patients reported lower prevalences of many other primary symptoms, such as brain fog (71.1%), unrefreshing sleep (55.8%), memory problems (38.4%), feeling of weakness (39.8%), sore/painful muscles (25.8%), and insomnia (34.8%). Furthermore, compared to Cluster 3, patients in Cluster 2 reported higher rates of certain comorbidities, including other dysautonomia (50.5%), MCAS (42.6%), EDS (33.5%), and craniocervical instability (16.0%).

Cluster 3: Cognitive and Sleep Dysfunction with Increased Pain. The functional capacity level of patients in this cluster is $43.8\% \pm 17.3\%$. Patients in Cluster 3 reported significantly higher percentages of brain fog (91.9%), unrefreshing sleep (85.5%), memory problems (73.5%), feeling of weakness (66.3%), sore/painful muscles (61.3%), and insomnia (54.1%) than those in Cluster 2. However, they reported markedly low rates of POTS as a primary symptom (3.7%) or a comorbidity (8.8%).

Cluster 4: Milder Symptomatology. This cluster included patients with the mildest symptoms and the highest mean capacity scores ($57.5\% \pm 20.6\%$), indicating relatively preserved functionality. Compared to other clusters, patients of this group exhibited lower average severity across almost all symptoms and comorbidities, including low rates of POTS (4.0% as a symptom and 15.3% as a comorbidity).

Separating respondents into those with ME/CFS and long COVID had minimal effect on the distribution of symptoms within each of the four clusters (*SI Appendix, Fig. S6*).

Perceived Effectiveness of Specific Treatments Differs Between Patient Subgroups. After defining separate symptom clusters, we next investigated whether survey respondents in the different clusters reported different responses to particular treatments. Fig. 4B shows that the treatments identified as having the greatest perceived benefits above—pacing and fluids/electrolytes—are effective across all the patient clusters. This is also consistent with the broad effect of these treatments on multiple core symptoms, as described above (Fig. 2B).

However, other treatments showed varying effects across different clusters of patients. In Cluster 1, which consisted of patients with the most symptoms and comorbidities, IgG (73.3%) and manual lymphatic drainage (73.9%) had the highest positive response rates, together with Fluids/Electrolytes (73.4%) and pacing (67.5%). In Cluster 2, where patients had POTS-dominant presentation, pacing (78.5%), fluids/electrolytes (69.3%), maraviroc (65.4%), and compression stockings (63.0%) received the highest NASs. For patients in Cluster 3, who experienced cognitive and sleep dysfunction and increased pain, ADHD Rx medications (62.1%) were reported as beneficial, in addition to pacing

(76.6%). However, patients in Cluster 1 reported no significant improvement from ADHD Rx medications to the reference. In Cluster 4, patients who had milder symptoms reported pacing (78.6%) and fluids/electrolytes (66.7%) as the most effective.

These findings suggest that understanding a patient's specific symptom profile may enable clinicians to tailor more effective therapeutic strategies.

Discussion

Our study uncovers striking similarities in the demographics, symptom presentation, comorbidities, and treatment outcomes between long COVID and ME/CFS patients. These findings highlight opportunities for integrative research on disease mechanisms and clinical care for both conditions (2).

Long COVID and ME/CFS patients in the study showed similar age and sex distribution, in addition to sharing most of the same symptoms and comorbidities, with few exceptions (Fig. 1). Both groups reported the same top four most frequent symptoms: fatigue, PEM, brain fog, and unrefreshing sleep, as well as high rates of OI-related symptoms, which match IOM ME/CFS diagnostic criteria(1). Other symptoms—memory problems, insomnia, dyspnea, headache, sore/painful muscles, numbness, and tingling—and comorbidities like anxiety and depression, POTS, migraine, other dysautonomia, EDS/joint hypermobility, MCAS, and ADD/ADHD, were also commonly present in both conditions. Importantly, responses to the broad range of treatments were also highly correlated between ME/CFS and long COVID patients (Fig. 3). Taken together, these findings are part of mounting evidence indicating overlapping pathologies of these illnesses (3, 7–9, 15, 44).

Many of the treatments among the top 20 (Fig. 2A) are supported by consensus guidelines (22–24), supporting the utility of this approach for assessing real-world evidence of therapies and identifying promising candidates for clinical trials. For example, pacing is recommended care to manage energy levels and avoid PEM in ME/CFS; fluid/electrolytes, compression stockings and Mestinon to help manage OI; beta-blockers and ivabradine to aid tachycardia present in POTS; IgG for use in immune dysfunction; antihistamines (H1 and/or H2 inhibitors) and other select treatments to manage MCAS; low-dose naltrexone to manage pain; and melatonin to mitigate sleep problems.

Other treatments in the top 20 provide evidence for considering these therapies when managing patient symptoms. For example, vitamin B12 deficiency is known to cause fatigue and other ME/CFS-like symptoms(22, 45), and B12 injections significantly outperformed oral B12 in helping patients, consistent with a previous study(46). Similarly, higher doses of coenzyme Q10 were also found to be significant, consistent with a recent trial in long COVID (47). Both CoQ10 and B12 improved fatigue, brain fog, and PEM, but not POTS. PEA, which has been investigated for pain control and reducing neuroinflammation (48), improved fatigue and brain fog. In addition, vagal nerve stimulation and manual lymphatic drainage helped multiple symptoms in patients with either condition.

Thrombotic sequelae and microclotting have been investigated in long COVID (3, 30–32). Nattokinase and lumbrokinase (NK/LK) have become popular fibrinolytic supplements due to their abilities to affect coagulation parameters. In this study, NK/LK and Rx anticoagulants/Rx antiplatelets were both shown to help significantly with multiple symptoms in both ME/CFS and long COVID, and remarkably, NK/LK yielded similar overall response as Rx anticoagulants/Rx antiplatelets. Of note, 5% of patients

reported discontinuing treatment of Rx anticoagulants/Rx antiplatelets due to bleeding concerns, including bruising, nose bleeds, bleeding gums, and heavier menstrual periods; one patient reported a “minor gastrointestinal bleed”.

Identifying specific patient subsets is crucial for effective treatment. Four clusters of patients were identified in this study, characterized by 1) multisystemic symptomatology, 2) POTS-dominant presentation, 3) cognitive and sleep dysfunction with increased pain, and 4) milder symptomatology (Fig. 4A). While pacing and fluids/electrolytes produced improvement across all clusters, specific treatments had the highest positive response rate in particular clusters (Fig. 4B). For example, ADHD Rx medications were most effective in Cluster 3, while they were no better than the reference in Cluster 1. Our findings represent an early effort to identify traits of patients most likely to benefit or be harmed by medications, which can eventually assist in establishing effective criteria or algorithms to determine which patients may benefit in clinical settings, i.e., precision medicine.

Clinical trials in ME/CFS and long COVID are urgently needed. Therapeutic candidates for randomized controlled trials (RCTs) for ME/CFS are typically identified through a combination of anecdotal patient self-reports, clinical observations, chart reviews, case reports, and open-label clinical trials (OLCTs) (49–53). This study represents an initial iteration of an approach that utilizes patient-driven surveys to assess outcomes of a large number of treatments, aiming to identify promising candidates for RCTs and specific patient subsets that may benefit from these treatments. Online treatment surveys can reach a wider, more diverse patient population than is typically feasible with other approaches. For example, case reports and OLCTs are typically limited to single or few specialty clinics where patients tend to share referral-driven characteristics and are well enough to reach the clinic. Because of the absence of objective biomarkers for long COVID or ME/CFS, most treatment studies to date are restricted to patient-reported outcomes. Large-scale survey studies that collect patient-reported treatment outcomes together with patients' demographic and clinical characteristics—such as comorbidities, clinical signs, symptoms, and disease severity—are particularly effective at determining whether certain treatments work for specific subsets of patients. The findings from this study support this approach. Considering the significant diversity among patients with these complex disease conditions, the development of clinical trials focused on targeted treatments may prove to be the most impactful strategy overall.

Our study has several important limitations. First, patients in the study were self-entered, and the effects of treatments were self-reported. While an online patient survey can reach a larger number of patients from a more diverse patient population, data collected are likely more prone to noise compared to case reports or OLCTs, where health professionals or researchers record treatment details. Future studies can benefit from recent advancements in Large Language Models (LLMs) and Natural Language Processing (NLP). Leveraging these technologies to analyze electronic health records (EHRs) can help establish ME/CFS and long COVID diagnoses, in addition to ICD-9 and ICD-10 codes. Additionally, NLP and LLMs can provide important information in identifying disease severity, symptoms, other clinical characteristics, and treatments of the patients. Second, patient surveys are subject to several potential biases, including sampling bias, e.g., selection bias and nonresponse bias, and response bias, e.g., recall bias (54, 55). Patient-based questionnaires have been developed and validated to evaluate symptom severity, functional limitations, and quality of life of chronic

conditions (56, 57). The questionnaire and the results of this study should be further verified. Third, due to the lack of randomization, confounders can affect the results of our study, similar to those found in case reports and OLCs. For example, treatment outcomes may be influenced by the placebo effect, which can vary based on the expectation of outcomes from specific treatments, and some patients may experience spontaneous improvement. Besides, patients might trial multiple treatments at once, making it difficult to attribute changes in patient conditions to a particular treatment. Further, patients had multiple symptoms and comorbidities, which adds to the complexity of evaluating the impacts of treatments on the disease as well as on specific symptoms. In this study, we attempted to offset some of these influences using Vitamin C (oral, nonliposomal) as a reference group. Fourth, although the current study included nearly 4,000 patients, many treatments still have fewer than 100 responses, which limits the reliability of the estimates of treatment outcomes. Further targeted surveys are needed to confirm these results. By utilizing prospective surveys, health-tracking apps, EHR integration, and increased patient education, future studies will allow us to extend our findings while increasing and verifying the accuracy of collected data.

In conclusion, this research establishes a foundation for utilizing patient-reported experiences from large-scale treatment surveys to provide real-world evidence for therapies in patient care and to inform candidates for future clinical trials in ME/CFS and long COVID.

Materials and Methods

Design of the TREATME Survey. The TREATME questionnaire was designed to survey individual experiences with a diverse spectrum of treatments used by ME/CFS and long COVID patients. The survey is available online (at https://www.surveymonkey.com/r/treatme_v2). Briefly, the first section of the survey includes questions on patient demographics, diagnoses, illness duration and severity, symptoms, comorbidities, and lab tests, while the remaining sections focus on treatments. Surveyed treatments were selected based on pharmacological and nonpharmacological therapies used in ME/CFS (25–27), emerging and pertinent clinical studies of long COVID, ME/CFS, and related comorbidities, and extensive discussions with patient community members. For example, antiplatelets, anticoagulants, and fibrinolytic supplements (e.g., nattokinase or lumbrokinase) were surveyed in part based on data suggesting platelet hyperactivation and other prothrombotic changes in long COVID patients (30–32), and other literature suggests the presence of coagulopathies in ME/CFS as well (33–35). Guanfacine, in combination with N-acetylcysteine, was surveyed based on neurobiological data and case reports showing cognitive benefits in long COVID (58). Orally administered nonliposomal vitamin C served as a comparator agent against which all other treatments could be measured. The use of oral vitamin C in this way was motivated by its widespread use, limited oral absorption (59), and inability to reach anywhere near the supratherapeutic concentrations which have shown promise in EBV “CFS” patients (60) or acute COVID inpatients (61).

In total, the survey included over 440 questions on the use and perceived efficacy of over 150 medications, supplements, and other interventions organized into 25 broader treatment categories. To borrow strength from treatments with

from those same seven options to evaluate a select treatment's effects on their most troubling symptoms; i.e., symptoms individually selected near the survey's beginning and later piped into specific treatment questions. Patients were also asked about side effects, duration of treatment, length of time before perceived benefit (if any), and duration of benefit (if present). When particularly pertinent, various questions about dosage, specific formulations, and supplement brands also were included. Patients were able to pause and resume the survey throughout the process and as many times as needed; the progress on the survey was stored in each respondent's web browser.

Contributions from the Patient Community in the Development of the Survey. Due to difficulties accessing appropriate medical care or effective treatments, many long COVID and ME/CFS patients suffering from these conditions have turned to social media to exchange information, discuss treatments, and even run their own patient-led experiments. Our survey grew out of discussions with a large cross-section of these individuals. Over a year prior to the creation of this survey, its author began releasing smaller surveys and polls on X/Twitter, gathering information on the efficacy of popular or controversial treatments used in the patient community. As the surveys gained traction, an increasing number of respondents provided suggestions and feedback which ultimately were utilized in the development of this survey.

Before the survey was released publicly, over 20 individuals with long COVID and/or ME/CFS took the survey as a trial run so that further feedback could be collected. Some expressed worries about the survey's length causing fatigue or PEM, so optional breaks were included throughout the survey, and both skip logic and advanced piping were employed to limit question burden. Trial respondents also stressed the importance of including “slightly” better/worse response options (rather than only “moderately” and “much” better/worse, which also had been considered), because even a “slight” improvement in symptoms from a given treatment made a significant impact on their quality of life, and oftentimes, improvements greater than “slight” were never reached. Thus, including the response option of “slightly” better/worse ensured a greater breadth of information was collected. Furthermore, it was also noted that sometimes treatments would work well for a while, but then inexplicably stop working, so a recurring question was added to clarify whether benefits were sustained or temporary. Such feedback from the community proved invaluable for survey construction.

Data Collection of the TREATME Survey. The study was reviewed and deemed exempt by the Kentucky Community and Technical College System (KCTCS) Human Research Protection Program (FWA00003332). Informed consent was obtained from all participants. Data were collected through an online survey (on [surveymonkey.com](https://www.surveymonkey.com)), which was distributed to patients with either condition through patient communities on Meta/Facebook, X/Twitter, Reddit, and StudyPages, from February 2023 to July 2023 (round 1) and October 2023 to February 2024 (round 2). 5,451 responses were received. Responses without a confirming answer on either ME/CFS or long COVID diagnosis (130) and/or those without answering any questions on treatments (1,295) were excluded from further analysis. For participants who responded in both rounds of the survey (101), only their first-round responses were retained for analysis. These resulted in 3,925 responses for downstream analysis, comprising 2,125 patients with ME/CFS and 1,800 with long COVID.

Evaluation of the Effect of Treatments from Patients' Responses. To evaluate patient-reported effects of treatments, we defined a NAS based on patients' responses on the impact of each treatment on their overall condition:

$$\text{Net Assessment Score} = \frac{(\text{Positive Assessments} - \text{Negative Assessments}) \times 100}{\text{Total Number of Assessments}}$$

similar modes of action, these treatments were consolidated into 97 treatment groups for our final analysis. To encourage more accurate results, patients were asked to review treatments only if they had a good idea of their effects, whether positive, negative, or neutral. Patients reported how they felt each treatment influenced their overall condition, choosing from seven options: much better, moderately better, slightly better, about the same/unchanged, slightly worse, moderately worse, or much worse. Additionally, patients were asked to choose

The number of positive and negative feedback corresponds to patients reporting any level of improvement or worsening, respectively. The total number of respondents reflects those who have tried the treatment and contributed feedback.

We also calculated the NAS of a treatment specific to the impact on each core symptom: fatigue or low energy, feeling worse after normal exertion (PEM), POTS, brain fog, and unrefreshing sleep.

For each treatment with a sample size of at least 20, the NAS for the overall condition or a specific symptom was then compared to that provided by patients for a Vitamin C supplement (oral, nonliposomal) as a reference.

The difference between the treatment and the reference group (vitamin C; oral, NOT liposomal) was analyzed using a two proportion Z test (62), which compares the proportion of improvement reported in each group. The test was applied as follows:

$$z = \frac{p_1 - p_2}{\sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where p_1 and p_2 represent the proportions of reported improvement (NAS) in the treatment and vitamin C reference groups, respectively, P is the combined proportion of improvement, and n_1 and n_2 are the sample sizes. The test statistic z follows a standard normal distribution, and the one-sided P -value indicates the probability of observing a proportion of improvement at least as extreme as p_1 under the null hypothesis that the treatment is not better than the vitamin C reference group. Adjusted P -values were controlled using the Benjamini-Hochberg procedure to adjust for multiple tests across various symptom reports, and a significant effect is defined as adjusted P -value (adj. P) < 0.05 .

Comparison of Treatment Response Between ME/CFS and long COVID.

To identify treatments with significantly different effects between ME/CFS and long COVID patients, a two-proportion Z-test was performed on the NAS of the treatment in the two conditions and the fold change between the scores. Only treatments taken by both ME/CFS and long COVID patients with a total sample size of over 100 were included in this analysis to reduce the impact of random effects on treatments with limited feedback. Treatments meeting abs (fold change) > 1.25 and multiple tests adjusted P -value < 0.05 were identified as significant between the two conditions.

To compare the impact of different factors on reported treatment efficacy, significant predictors of treatment effectiveness (i.e., the NAS) were identified

from disease diagnosis (ME/CFS vs long COVID), severity (patient capacity level), and demographics of patients. A gradient boosting machine (GBM) model (63) was employed, utilizing a Gaussian loss function. It included 5,000 trees and was validated using 5-fold cross-validation.

Subclustering of Patients Based on Symptoms and Comorbidities. Uniform manifold approximation and projection (UMAP) (64, 65) was used to transform the high-dimensional patient data, encompassing demographic information, symptoms, and comorbidities into a two-dimensional space. K-means clustering was then applied to cluster patients into subgroups. The optimal number of clusters was determined as 4. For each treatment with a sample size greater than 20 in a cluster, the NAS for that treatment was calculated within the cluster.

Data, Materials, and Software Availability. The survey and the detailed results are available in [Datasets S1–S3](#).

ACKNOWLEDGMENTS. The survey grew out of discussions with many patients in the ME/CFS and long COVID communities, and numerous patients provided invaluable feedback throughout the project. We would also like to express our sincere gratitude to the more than 5,000 patients who participated in this study. This study would not have succeeded without the tremendous support from the patient communities. We would also like to extend a special thank you to Dr. Trent Garrison, who helped Dr. Eckey with critical steps behind the scenes. Additionally, we appreciate the helpful advice we received from physicians specializing in ME/CFS and long COVID. Dr. Donna Felsenstein helped review the manuscript. This work was partially supported by the Open Medicine Foundation (WX).

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